

Grover's Algorithm

Given a Boolean function $f: \{00, 01, 10, 11\} \rightarrow \{0, 1\}$ with only one input ω satisfies the condition $f(x) = 1$.

What is omega ?

The two qubits q_1q_2 are initiated in the state 00 .

A Hadamard gate $\left(H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)$ to make a superposition.

$$\begin{aligned} H^{\otimes 2}|00\rangle &= (H \otimes H)|0\rangle \otimes |0\rangle = H|0\rangle \otimes H|0\rangle \\ &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = |S\rangle \end{aligned}$$

Now we define the oracle.

$$U_{\omega}|x\rangle = (I - 2|\omega\rangle\langle\omega|)|x\rangle = (-1)^{f(x)}|x\rangle = \begin{cases} -|x\rangle; & x = \omega \\ |x\rangle; & x \neq \omega \end{cases}$$

Here, we assume it's the 11 state that verifies the condition.

$$U_{\omega}|S\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle) = |S'\rangle$$

Now, we define the Grover diffusion operator.

$$D = 2|S\rangle\langle S| - I$$

Finally, we get our omega state.

$$DU_{\omega}|S\rangle = |\omega\rangle$$

